

PCB technologies eliminate the need for an external plate by integrating a metal core layer directly into the board that allows the PCB itself to act as a heat spreader.

Heat pipes. An extremely efficient method of thermal conduction is provided by heat pipes. These devices consist of a sealed, hollow tube containing a liquid or coolant. One end of the tube is attached to a heat source and the other to a secondary heat exchanger, such as a heat-sink. Heat generated from a component boils the liquid at one end of the pipe, which then travels to the cool end. The vapour condenses (giving off latent heat) and returns back to the heated end through capillary action or gravity. Heat pipes are typically made of a thermally conductive metal, which can be flat or bent into complex shapes. This makes these very suitable for space-limited board designs.

Thermal interface materials. An important means of optimising heat transfer between a heat source and a cooling device (or between two adjacent cooling devices) is the use of thermal interface materials. These are thermally conductive, pre-manufactured materials that can be applied in a variety of forms. These include thermal pads, adhesives, gels and grease. These are designed to conform to surface irregularities and imperfections, which allow these to fill in air gaps between contact surfaces. This ensures that maximum available surface area is being used for heat transfer.

With heat-generating circuitry it may be adequate to ensure thermal control by potting into a metal box, with or without integral or attached cooling fins, using a thermally-conductive potting compound.

Heat transfer compounds are designed to fill the gap between the device and the heat-sink and, thus, reduce the thermal resistance at the boundary between the two. This leads to faster heat loss to the heat-sink and a lower operating temperature for the device.

Heat transfer compounds can be of various types. Electrolubes produce a range of thermally-conductive pastes, which can be applied to displace

air from the component/heat-sink interface. These consist of thermally-conductive mineral fillers in a carrier fluid—the fluid may be non-silicone or silicone based.

Silicone-based pastes generally have higher operating temperatures than their non-silicone alternatives. Silicones can cause a problem with some circuitry as these migrate very readily and can cause contamination on, say, relay contacts. It is possible to modify the thermal conductivity of thermally-conductive pastes by



Fig. 3: Using heat pipes in electronics cooling (Credit: www.atlastsolutions.com)



Fig. 4: Cold plate (Credit: www.lytron.com)

increasing the mineral filler content, or changing the types of filler used. These have a higher filler loading and contain a special blend of different fillers to maximise thermal conductivity.

Generally, higher thermally-conductive pastes are higher in viscosity. This can lead to greater difficulty in dispensing. Thermally-conductive pastes remain as pastes. This makes for easier disassembling of components for recovery or repair. It may be desirable in some circumstances to use a thermal transfer material that cures up to a solid. With any thermally-conductive material it is very important to ensure that the interface between the device and the heat-sink is completely filled and all air is displaced.

Cold plate. Placing a conductive thick metal plate, referred to as a

cold plate, as a heat transfer interface between a heat source and a cold flowing fluid (or any other heat-sink) may improve cooling performance. In such an arrangement, the heat source is cooled under the thick plate instead of being cooled in direct contact with the cooling fluid. A thick plate can significantly improve the heat transfer between the heat source and the cooling fluid by way of conducting the heat current in an optimal manner.

The two most attractive advantages of this method are no additional pumping power and no extra heat transfer surface area. This is quite different from fins (extended surfaces).

Forced air. The most common method of active cooling is the application of forced air over a heat source. This is accomplished by using a fan or blower to increase the flow of air directly above a heat-generating component or a passive cooling device, such as a heat-sink. The increased air flow moves heated air away at a faster rate. This significantly improves overall heat dissipation.

Fans of different sizes and configurations can be used to control the flow rate, while the flow path can be optimised by their proper placement and orientation. This is especially important in enclosed systems where heated air must be channelled to an exhaust vent. More complex systems utilise electronic fan control to manage both variables based on feedback from localised temperature sensors.

Another type of active heat management is pumped liquid cooling. This closed-loop system consists of a circulating liquid coolant moved by a pump through discrete tubing to remove heat from a heat source.

In electronic cooling, the most common application of this technology is cold plates. A cold plate is a thermally-conductive metal containing internal channels or embedded tubing, which allows liquid coolant flow. It is typically mounted on the back of a PCB containing heat-generating components. The heat source heats up the liquid coolant inside the cold plate. This is pumped to a secondary heat exchanger, where it is cooled. The cooled liquid is then returned to